
Waves



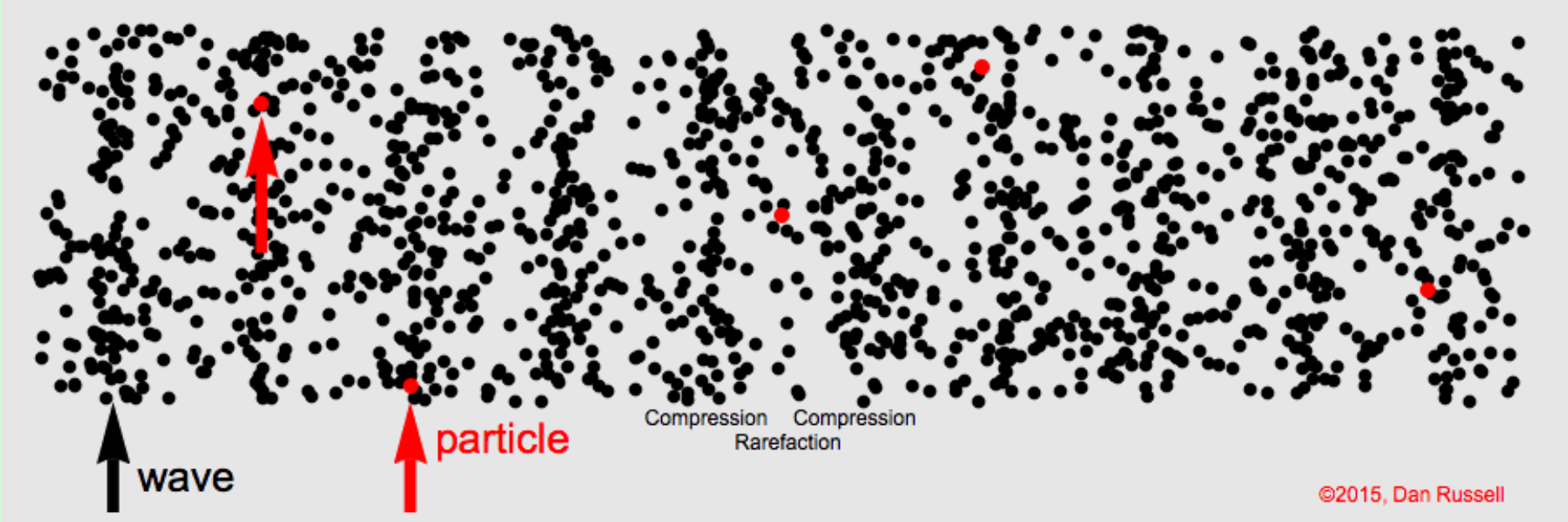
Heart Beat

Largest Wave Surfed

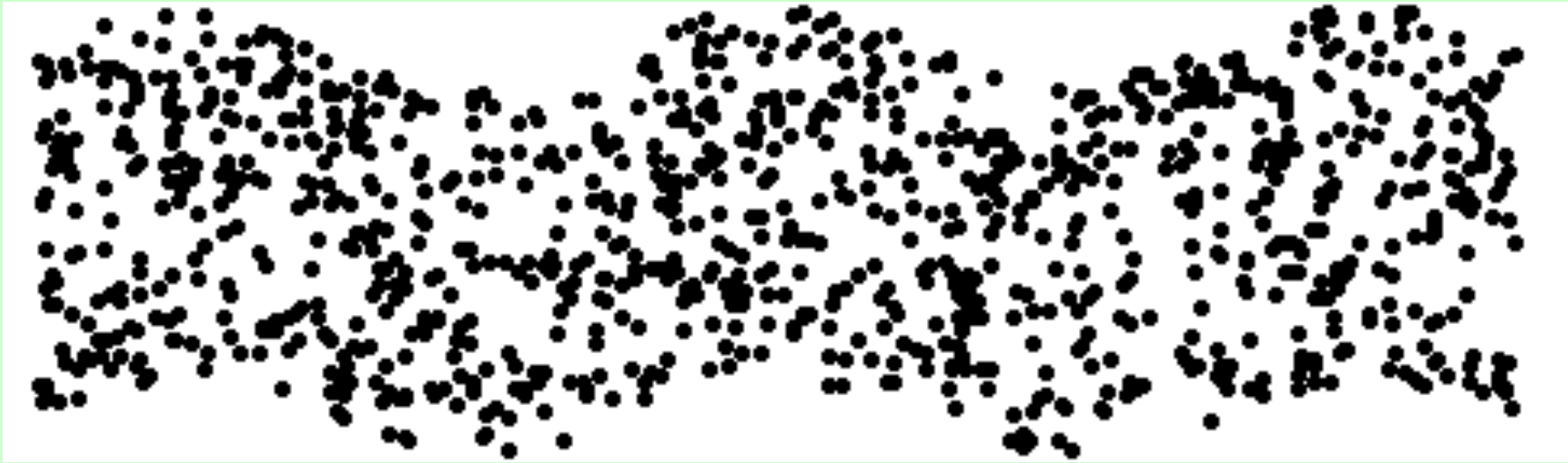




**Longitudinal Wave**: Particles oscillate in same direction as wave propagates

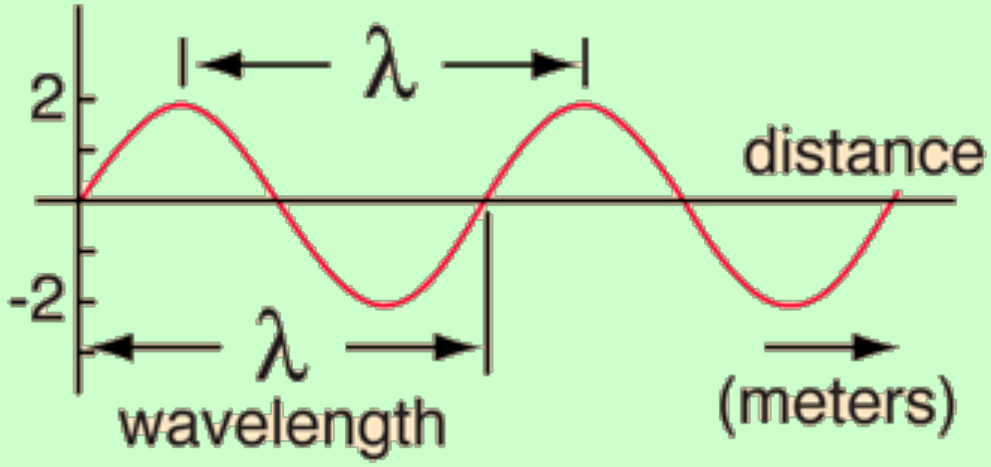


**Transverse Wave**: Particles oscillate perpendicular to  direction wave propagates

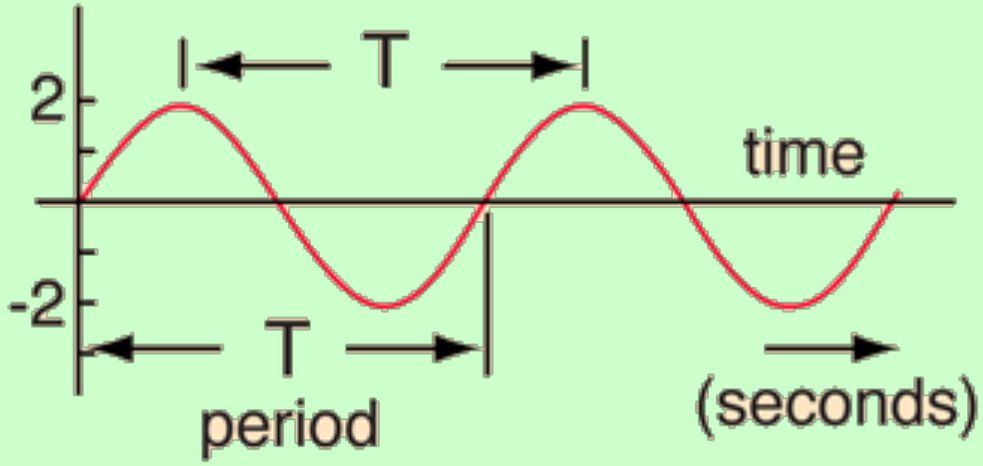


Transverse waves are actually two waves…

**Spatial wave:**
(snapshot)

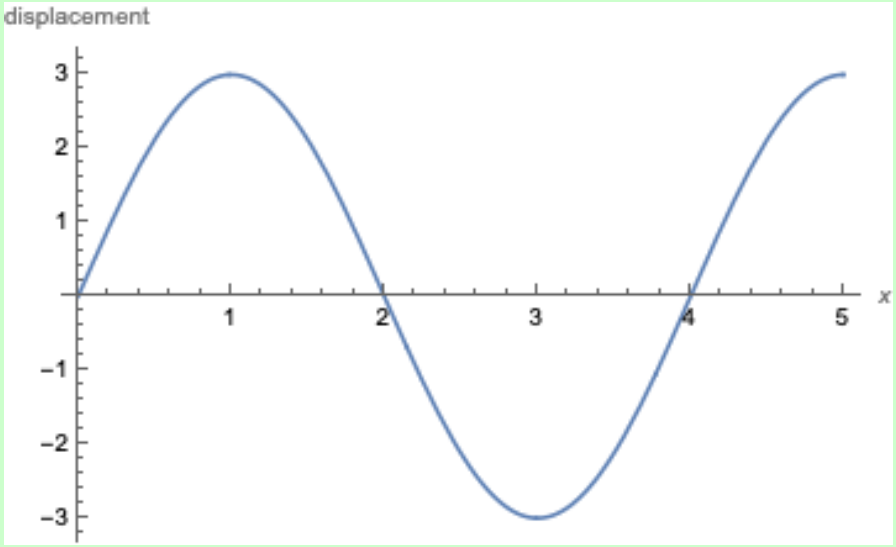



**Time wave:**
(history)

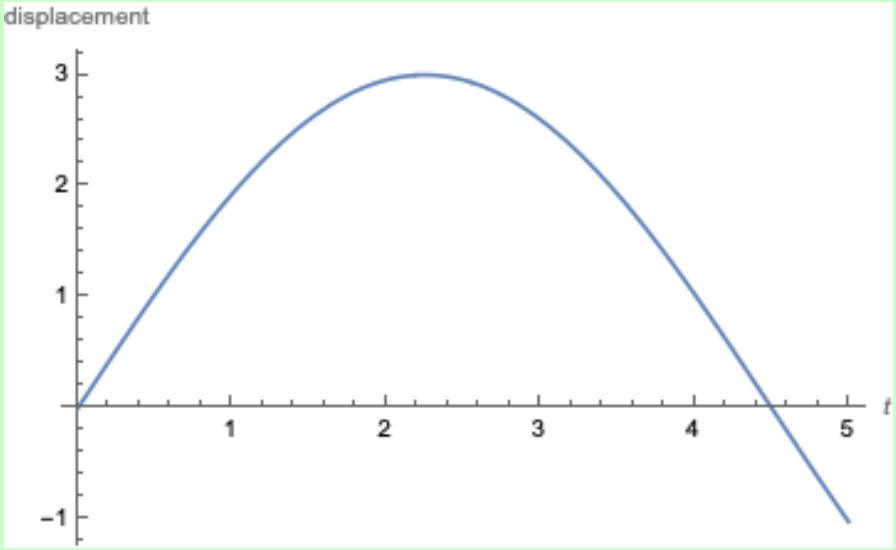


Ex. Find Amplitude (A), Wavelength ($\lambda$), and Period (T)

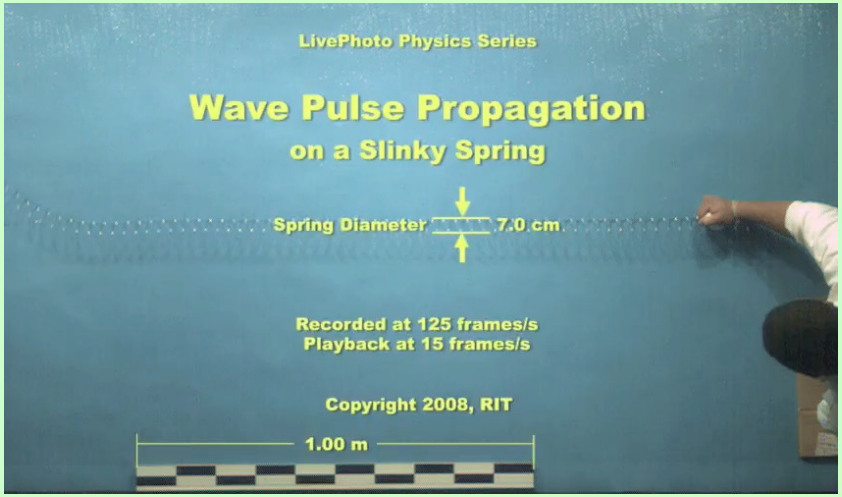
**Snapshot**



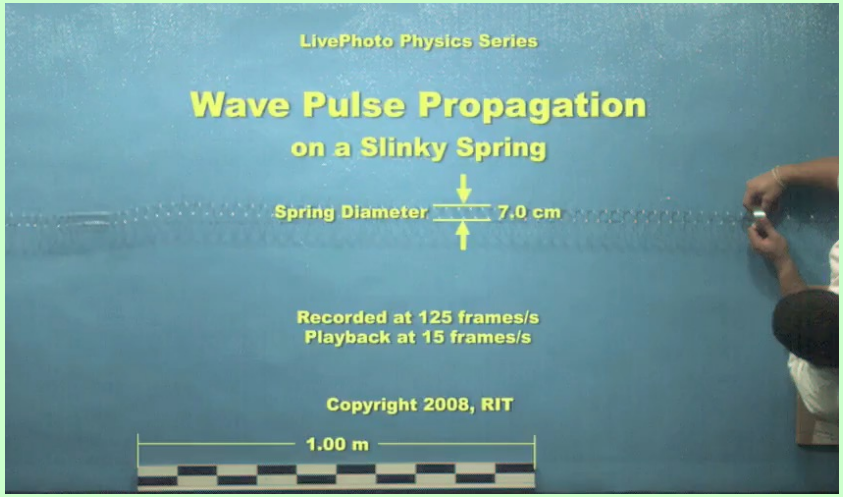

**history**



# Waves are built from pulses




Transverse Pulse




Longitudinal Pulse

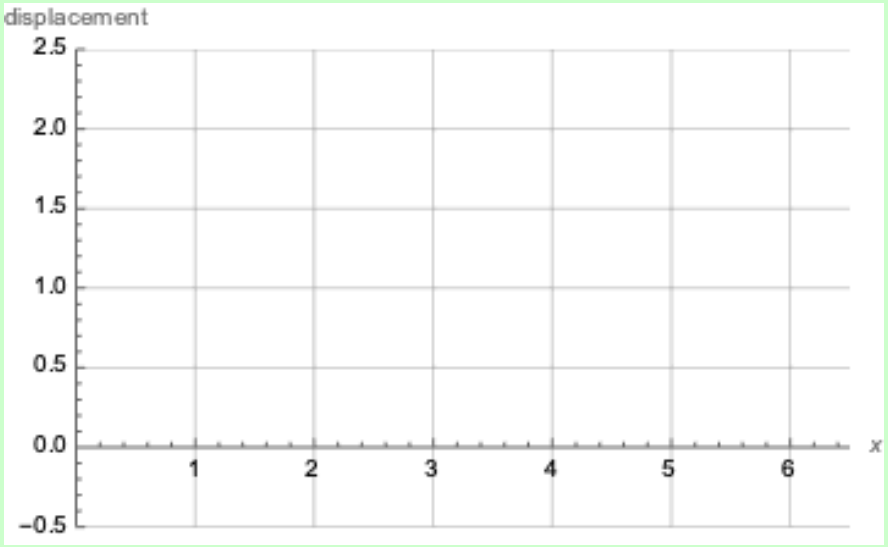
Slinky demo!!

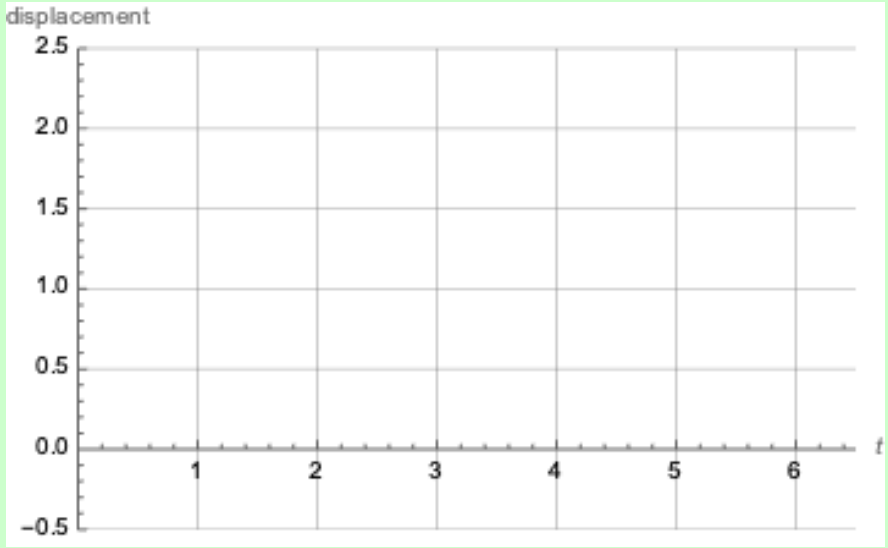
# EX. Pulses

Initial pulse d(x,0)



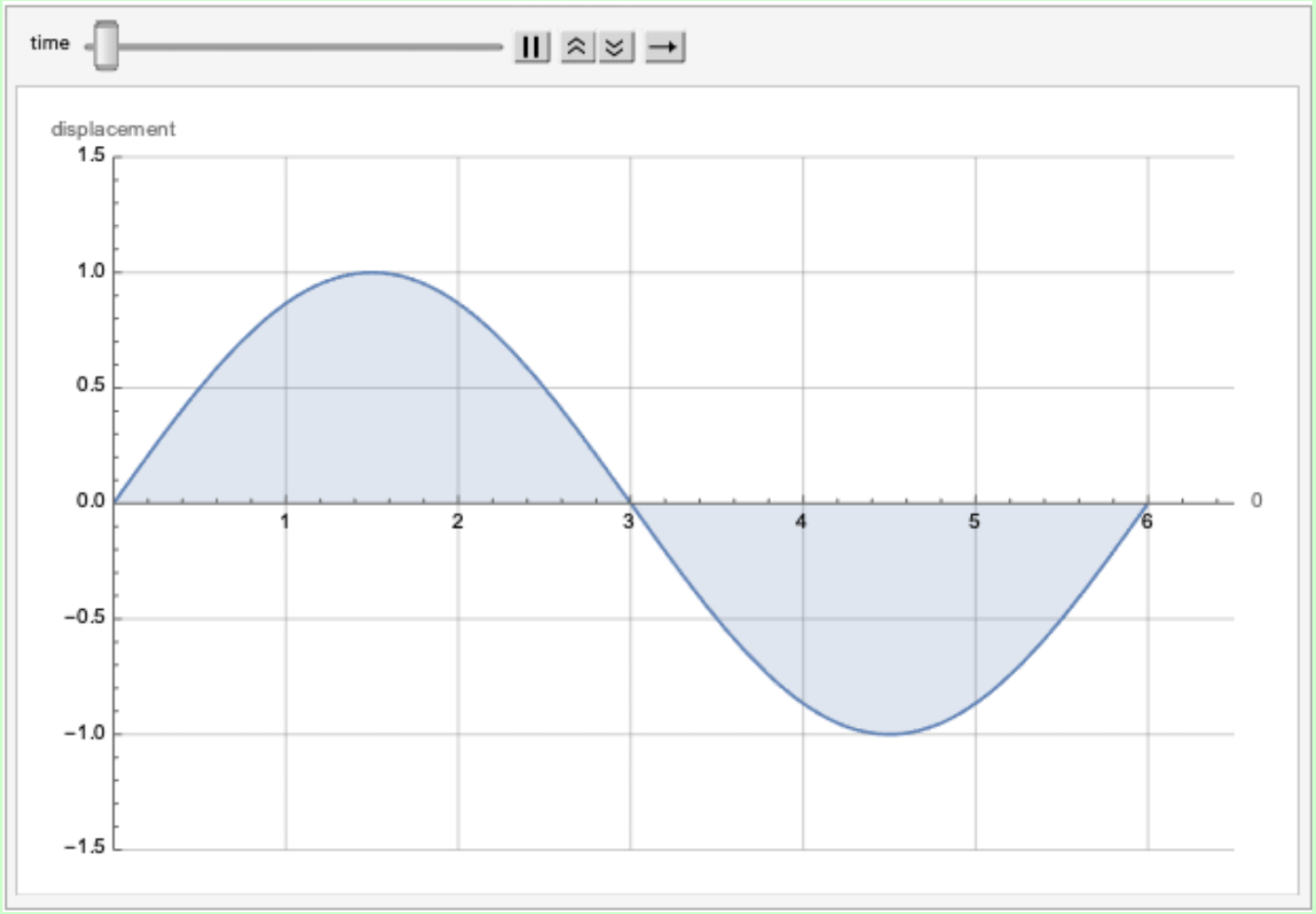

**1. Make snapshot: d(x,t=4)**




**2. Make snapshot: d(x=4,t)**

If I keep sending pulses a wave will move with some velocity




Estimate v

Graph the following:

1. $y = A\sin x, x\ \epsilon\ [0, 2\pi]$

2. $y = 3\sin(2x), x\ \epsilon\ [0, 2\pi]$

3. $y = 3\sin(x + \frac{\pi}{2}), x\ \epsilon\ [0, 2\pi]$

4. Make a graph with $\lambda = 3\pi$ , A = 3

5. Make a sin function that has arbitrary wavelength, amplitude, and phase constant

Compare to notes: Section 1.1: pg 3

Ex. Graph a Sine wave with:

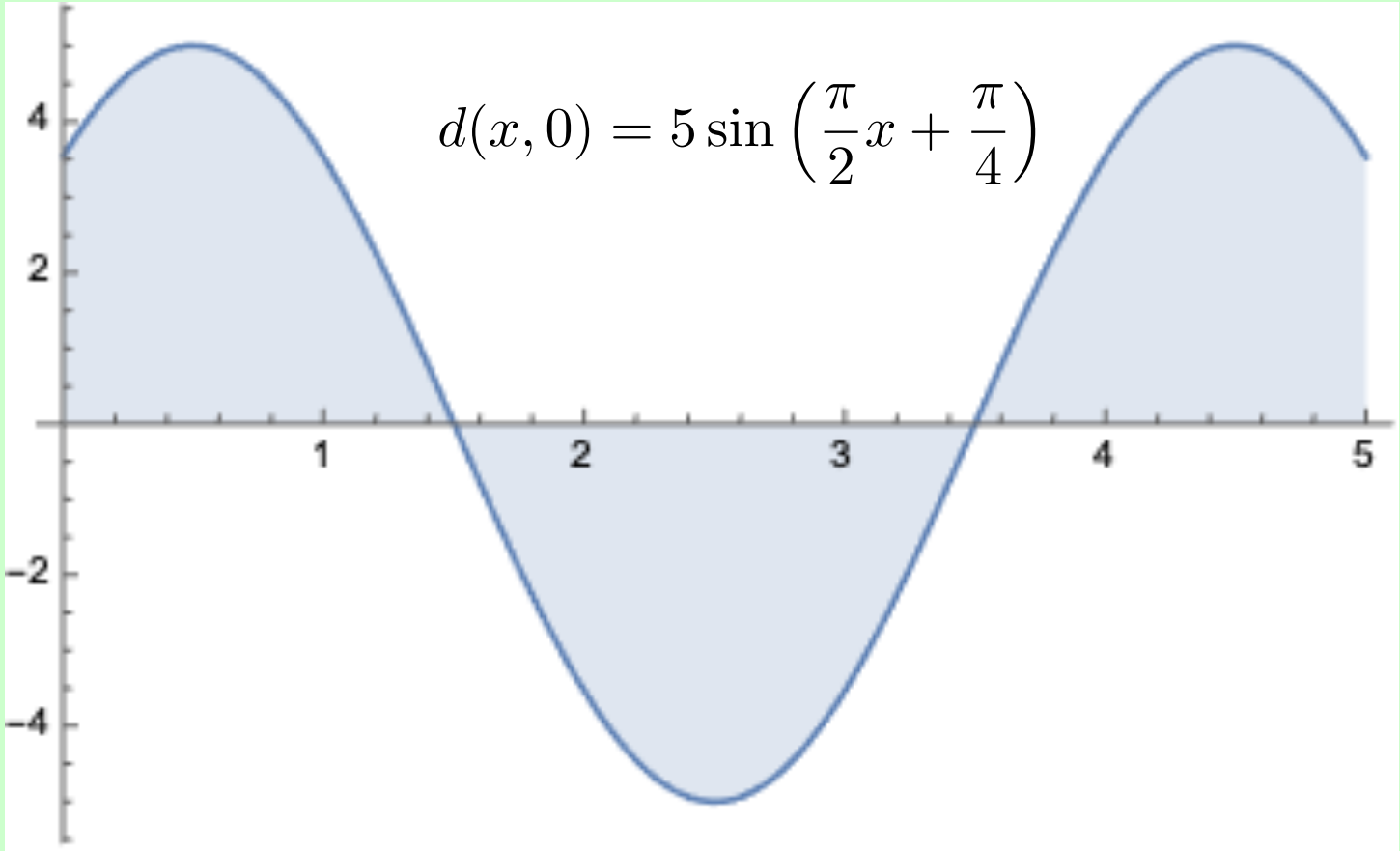
A = 5

$\lambda = 4$

$\phi_0 = \frac{\pi}{4}$

Snapshot eqn:

$$d(x,0) = A\sin\left(\frac{2\pi}{\lambda}x + \phi_0\right)$$



## Wavenumber, k

We can rewrite the snapshot eqn: $d(x,0) = A\sin\left(\frac{2\pi}{\lambda}x + \phi_0\right)$ ,

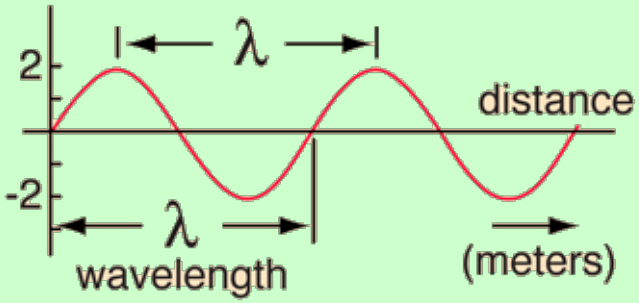
$$\boxed{d(x,0) = A\sin\left(kx + \phi_0\right)}$$

**k is called the Wavenumber, and has units of radians per m**
**It means how many radians of phase you scan through as you walk through 1 m of a wave.**

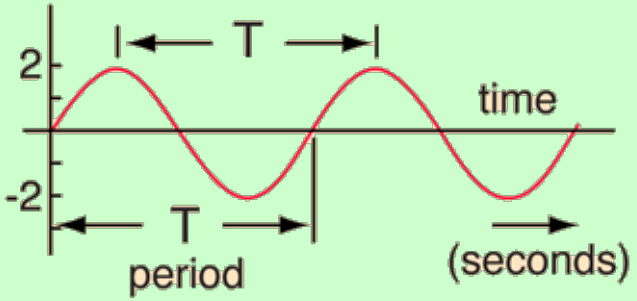
Ex. What is the wave number of H$\alpha$ $= 656.3\ nm$

## Wave velocity

$$v = \frac{d}{t} = \frac{\lambda}{T}$$




d




t

People don't like 1/T, so define frequency:

$$f = \frac{1}{T}$$

thus

$$\boxed{v = \lambda f}$$

(wave velocity eqn)

(Units Hertz, Hz = 1/secs)

Ex. What is the frequency of a 0.05 s wave?
If its wavelength is 3 m, what is its velocity?

**Derivation of Wave equation**

$$d(x,0) = A\sin(kx + \phi_0)$$

If wave moves @ const speed...

$$d(x,t) = d(x - vt, 0)$$

$$d(x,t) = A\sin\left(k(x - vt) + \phi_0\right)$$

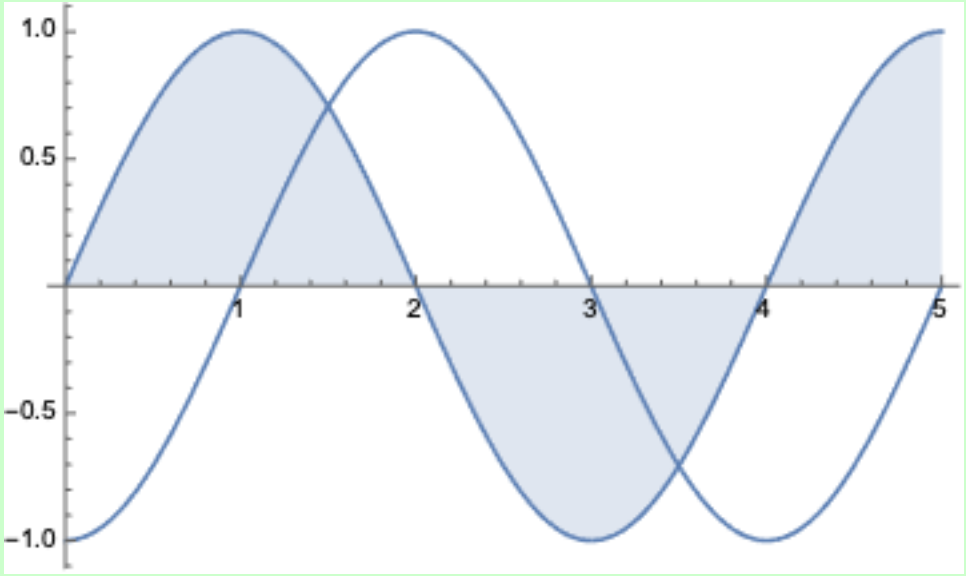
$$= A\sin\left(Kx - \frac{2\pi}{\lambda}vt + \phi_0\right)$$

$$= A\sin\left(Kx - \frac{2\pi}{\lambda}\lambda f t + \phi_0\right)$$

$$= A\sin\left(Kx - 2\pi f t + \phi_0\right)$$

$$d(x,t) = A\sin(kx - \omega t + \phi_0)$$

The wave has a velocity of 1 m/s




## Angular frequency, $\omega$

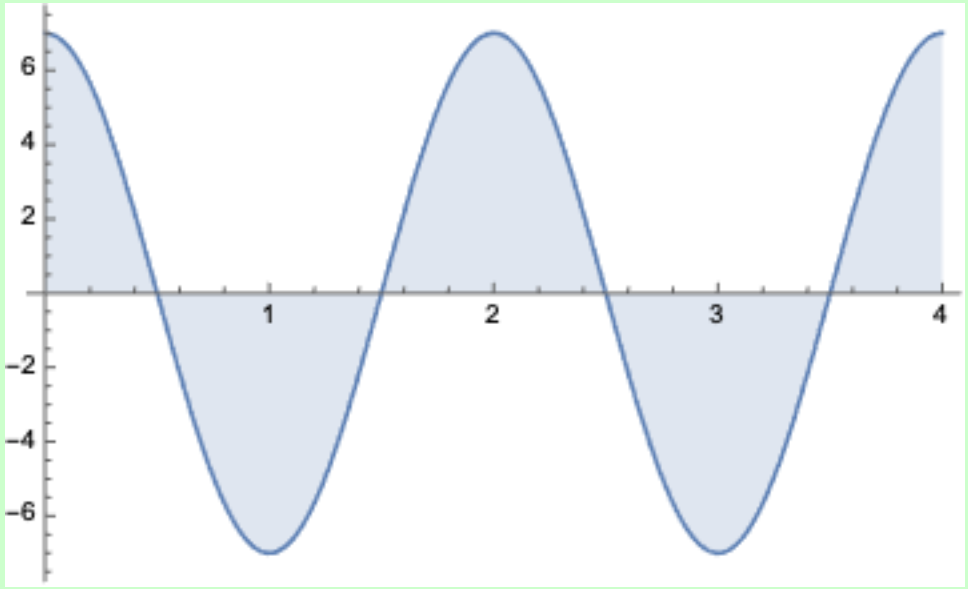
$$\omega = 2\pi f$$

Units: rad/s

Another short hand. It means the number of radians that flow through you In 1 second as a wave passes.

Ex. Find the following quantities for the wave depicted whose v = 200 m/s

1. f
2. T
3. $\omega$
4. $\lambda$
5. A
6. $\phi_0$




7. Determine d(x,t)
8. Sketch the wave at t = T/4

# Main Equations

$$d(x,t) = Asin(kx - \omega t + \phi_0)$$ (Wave eqn)

$$v = \lambda f$$ (Wave velocity)

$$f = \frac{1}{T}$$ Frequency (Hz)

$$\omega = 2\pi f$$ Angular freq (rad/s)

$$k = \frac{2\pi}{\lambda}$$ Wave number (rad/m)